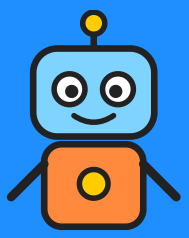
Robotics Project #12 · Kids.JoinDev

Line-Follower Robot

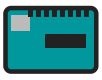
Ages 10+ · 2 hours · Build with a grown-up!



🎯 Goal

Build a wheeled robot that follows a black tape line all by itself.

🧰 You need



Arduino Uno



2 IR line sensors (TCRT5000)



2 TT gear motors + wheels



L298N or L9110 motor driver



4xAA battery pack



cardboard chassis + caster

🔧 Build it

1. Make a cardboard chassis and glue on the 2 motors and a caster. ☐
2. Mount the 2 sensors at the front, just above the floor, one each side. ☐
3. Wire sensors to pins 2 and 3, motor driver to pins 5, 6, 9, 10. ☐
4. Make a track with black tape on light floor. ☐
5. Upload line_follower.ino and put the robot on the line. ☐
6. Left sensor sees black? Turn left. Right sees black? Turn right! ☐

⚠️ Safety

Grown-up checks the battery wiring. Switch off before changing wires. Keep hair and fingers away from wheels.

💡 How it works

Black tape reflects less light. Each sensor says 'line' or 'no line', and the code steers to keep the line between them.
Code: line_follower.ino (in the project folder)